BoltzDesign1

Inverting an all-atom predictor for biomolecular binder design

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Computational proof of concept; no experimental binding validation.

Can one predictor design binders across target chemistries?

The design problem

Given a target, find an amino-acid sequence that folds and forms a suitable interface. Targets may be proteins, small molecules, DNA or modified residues.

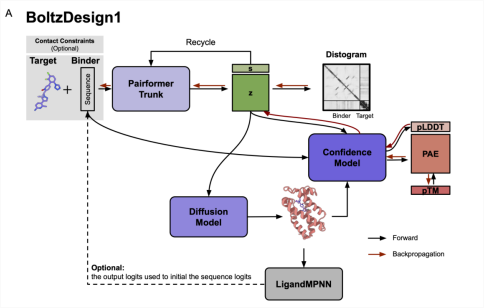
The opportunity

Boltz-1 already models these chemistries. Can we optimize its *input sequence* instead of training another generative model?

The bottleneck: differentiating through an AF3-like coordinate diffusion process is expensive. The paper sidesteps this path.

Introduction, pp. 1–2. The paper evaluates Boltz-1, not later Boltz generations.

The central move: optimize distances before coordinates



Pairformer predicts distributions over pairwise distances. Backpropagate a contact objective from these distributions to sequence logits. Optionally add confidence losses while stopping gradients through diffusion coordinates.

Fig. 1A; Methods Eq. 1. Original authors' diagram, cropped.

Inputs and outputs: a small-molecule running example

Example: a protein binder for the cofactor SAM
Input: SAM molecular identity, a chosen binder length, initial amino-acid logits, and optional contact constraints. No new model training.

1. Optimize an $L \times 20$ array of sequence logits with frozen Boltz-1 weights.
2. Convert the optimized sequence to amino acids; optionally redesign using LigandMPNN.
3. Repredict the complex using AF3 and evaluate confidence and consistency.

Output: candidate protein sequences and modeled complexes, not measured affinities or verified selectivity.

Methods; small-molecule experiment, p. 6; Fig. 3. SAM is one of four benchmark ligands.

Two gradient routes, one frozen structure predictor

For sequence parameters θ, let the trunk output single and pair features (s, z).

$$(s, z) = f_{\text{trunk}}(\theta, \text{target}), \qquad q_{ijm} = \text{softmax}_m(d_{ijm}).$$

1. **Pairformer only:** optimize contact losses computed from q. No coordinate diffusion is needed along this gradient route.
2. **Pairformer + Confidence:** sample coordinates x, then use

   $$c = f_{\text{conf}}(s, z, \text{stopgrad}(x), \theta).$$

   Gradients reach sequence through trunk and confidence features, but do not traverse the diffusion trajectory.

Methods, p. 2, Eq. 1; Fig. 1A. Diagrammatic notation simplified for teaching.

What does the contact objective ask for?

Each pair has 64 distance bins over the reported 2–22 Å range. Let q be the full distribution and q^* its renormalization restricted to bins below a cutoff:

$$\ell_{ij} = -\sum_m q^*_{ijm} \log q_{ijm}.$$

1. Low loss favors concentrated probability on contact-compatible distances.
2. Take the lowest k pair losses per binder position, then average the best l binder positions.
3. Default: $k = 1$ for inter-contact, $k = 2$ for intra-contact; $l = L$. Intra-contact excludes nearby sequence positions to encourage long-range packing.

Results, pp. 4–6, Eqs. 2–5. Cutoffs: intra 14 Å, inter 22 Å; these are optimization thresholds, not physical binding criteria.

From a soft sequence to a real sequence

1. **Warm up:** softmax logits at temperature 1 to avoid unrealistic raw-input behavior.
2. **Explore:** interpolate logits and softmax representations as optimization progresses.
3. **Sharpen:** anneal softmax temperature toward 0.01 and decay the learning rate.
4. **Discretize:** one-hot amino acids in the forward pass, with gradients through softmax (a straight-through estimator).

LigandMPNN can then redesign the sequence, or supply initialization. The design variables are sequence representations; model parameters remain fixed.

Methods, p. 4. Four optimization stages followed by optional LigandMPNN redesign.

A practical failure mode shapes the SAM objective

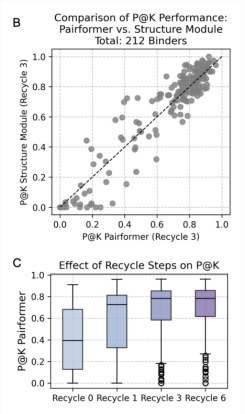
Why not maximize ligand contacts everywhere?
The authors observed proteins wrapping around a small molecule, with large apo–holo structural differences. A highly scored complex may depend on the ligand to hold the protein together.

1. Begin without the ligand so protein secondary structure can emerge.
2. Introduce the ligand after warm-up.
3. Optimize only the single best inter-contact at each step ($k = 1, l = 1$), allowing additional supporting contacts to emerge indirectly.

This is a task-specific heuristic; all-atom modeling alone does not solve objective design.

Small-molecule design protocol, p. 6. Apo–holo explanation is the authors' motivation for the heuristic.

Does the distogram retain useful interface information?



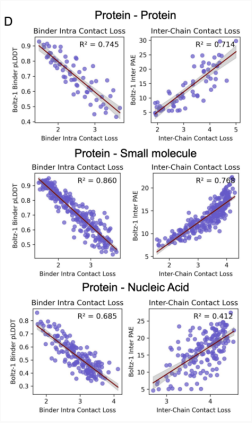
212 successful BindCraft binders across 13 targets. Reference contacts: AF2-predicted Cβ distances < 8 Å.

76% have distogram precision at $K > 0.5$, where K is the reference number of contacts.

Recycling improves contact prediction until about three recycles. This tests *existing designs*, not design optimization.

Fig. 2B–C; p. 4. Reference is a predicted model, not an experimental contact map.

Confidence proxies depend on target type



Intra-contact loss tracks binder pLDDT; inter-contact loss tracks interface pAE for protein and small-molecule designs.

For nucleic acids, inter-contact loss versus interface pAE has $R^2 = 0.412$: a weaker relationship.

Consequence: the authors favor additional confidence support for nucleic-acid interfaces. Contact certainty does not uniquely determine docking geometry.

Fig. 2D; p. 6. Correlations are within model-derived metrics; they do not calibrate binding probability.

Small-molecule benchmark: what counts as success?

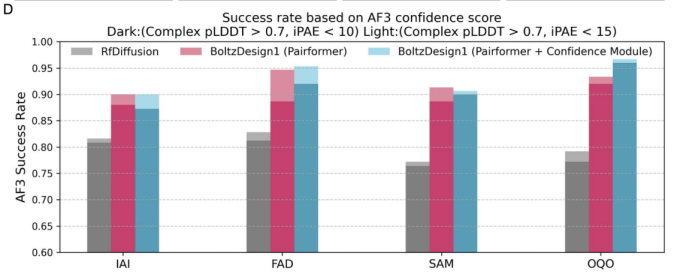
Four ligands from the RfDiffusionAA study: IAI, FAD, SAM and OQO.

1. Generate 30 structures per ligand, with helix-loss weights in the reported 0 to -0.3 range.
2. Use LigandMPNN to generate five sequences per structure; assess complexes using AF3.
3. Main filter: AF3 complex pLDDT > 0.7 and interface pAE $< 10\,\text{Å}$. A relaxed filter uses pAE $< 15\,\text{Å}$.

Other checks: cross-model and redesign self-consistency RMSD $< 2\,\text{Å}$, Gnina docking scores and pairwise TM-score diversity.

p. 6; Fig. 3D. pLDDT is reported on a 0–1 scale here. Five redesigns share each generated backbone.

AF3 pass rates improve across the four ligands



Both BoltzDesign1 routes exceed RfDiffusionAA in this benchmark. Adding Confidence improves the rate for three of four ligands.

Dark segments use the strict filter; light extensions show the relaxed filter. These bars measure predictor agreement and confidence, not laboratory success.

Fig. 3D; p. 8. Best protocol uses zero recycles and fixed interface residues during LigandMPNN redesign.

Which protocol choices matter?

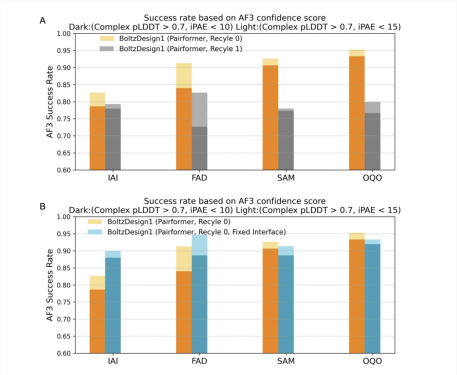


Figure S2: BoltzDesign1 achieves the highest success rate when run without recycling steps, redesigning sequences with LigandMPNN while keeping the initial BoltzDesign1 interface residues fixed. (A) Success rate of BoltzDesign1 (Pairformer) with recycle = 0 vs. recycle = 1. (B) Success rate of BoltzDesign1 (Pairformer) designs fully redesigned with LigandMPNN vs. designs

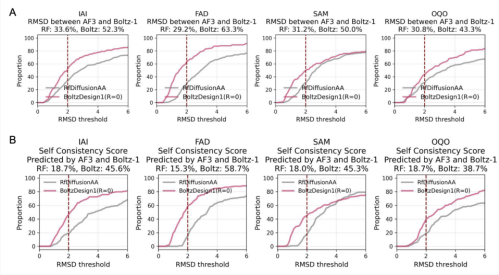
Zero versus one recycle: zero does better in design, despite recycling helping contact prediction.

Fixed interface: retain initial residues whose Cα lies within 6 Å of any ligand atom; redesign the rest.

This outperforms full redesign. The authors hypothesize that extra recycling can support adversarial optimization; this mechanism is not established.

Fig. S2A–B; Fig. S1; pp. 6, 8. Interface recovery by LigandMPNN is also higher than surface recovery.

Consistency supports designability, with a clear ceiling

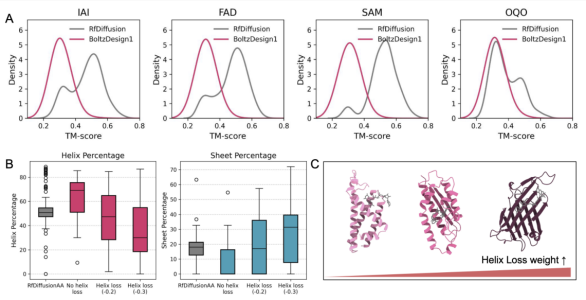


BoltzDesign1 has stronger AF3–Boltz-1 agreement and redesign self-consistency than the baseline (S3; Pairformer-only results in S4).

Gnina scores remain below the native complex for most designs. Fractions exceeding native scores: SAM 9.3%, OQO 7.3%, IAI 4.0% (S3C).

Figs. S3A–C, S4; p. 8. RMSD and docking scores are computational proxies, not measured fold stability or affinity.

The method can vary fold topology and secondary structure

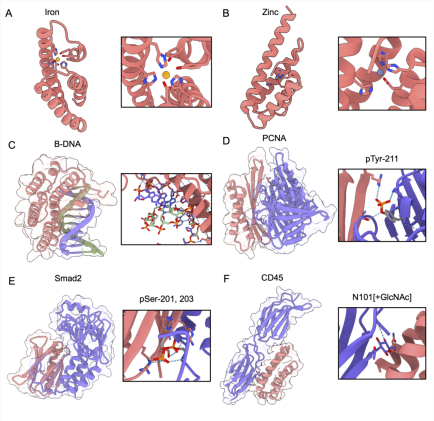


Unfiltered sets of 30 designs per ligand: average pairwise TM-score is 0.36 for BoltzDesign1 versus 0.46 for RfDiffusionAA. Lower similarity indicates more diverse modeled folds.

Increasing the magnitude of the negative helix-loss weight increases sheet content. Diversity was measured before success filtering.

Fig. 4A–C; p. 8. Diversity and high success rate are separate measurements.

Beyond small molecules: examples broaden the target space



Metals: iron and zinc designs, checked with AF3 and AllMetal3D.

B-DNA: modeled shape complementarity, positive interface and hydrogen bonds.

Modifications: phosphorylated PCNA and Smad2; glycosylated CD45.

These are structural demonstrations, not benchmarked success rates or experimentally verified specificity.

Fig. 5A–F; pp. 9–10. Site constraints and modification tokens are inputs.

Flexible ligand modeling is a capability, not yet a mechanism

1. RfDiffusionAA keeps ligand conformation fixed during the compared generation process.
2. BoltzDesign1 can sample a new ligand conformation as the sequence changes. This can explore coupled binder–ligand geometry.
3. The distogram route avoids differentiating through a specific sampled coordinate trajectory; confidence can still use sampled coordinates.

Our interpretation: this is useful search flexibility. The paper does not establish a thermodynamic induced-fit pathway, free-energy minimum, or conformational ensemble accuracy.

Introduction; Methods; small-molecule protocol and Conclusion. No physical free-energy benchmark is presented.

What the evidence establishes—and what it leaves open

Supported

Sequence optimization can bypass diffusion backpropagation.

Four-ligand AF3 filtering and structural diversity improve over the chosen baseline.

Multiple chemistries admit plausible modeled complexes.

Open

Expression, solubility, affinity, selectivity and functional effects.

Generalization across broader target sets; independent calibration of confidence.

Quantitative speed or memory gains under matched hardware and settings.

Results and Conclusion. Authors acknowledge no templates, no nucleic-acid MSA integration, and model-related overfitting risk.

A decisive next experiment would test specificity

Example follow-up: a phosphorylation-dependent binder
Blindly select candidates spanning confidence bins. Measure binding to the phosphorylated target, the unmodified target and unrelated phosphoproteins; check expression and monomeric folding independently.

1. Does modeled interface confidence predict any measurable binding?
2. Does encoding a modification generate recognition of that state, rather than generic electrostatic attraction?
3. Do zero-recycle and fixed-interface choices improve wet-lab yield under matched budgets?

Discussion proposal, not an experiment reported in the paper. Motivated by Fig. 5 and the stated experimental-validation gap.

Take-home message and sources

BoltzDesign1 turns an all-atom predictor into a sequence optimizer by moving the objective upstream of coordinate diffusion.

The strongest evidence is a four-ligand computational benchmark and associated ablations. The broader target examples motivate experiments; they do not demonstrate binding in cells or in vitro.

Cho et al. *BoltzDesign1: Inverting All-Atom Structure Prediction Model for Generalized Biomolecular Binder Design*. bioRxiv (2025), v1.
`https://doi.org/10.1101/2025.04.06.647261`

Full PDF includes supplementary Figs. S1–S4. Code:
`https://github.com/yehlincho/BoltzDesign1`

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